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QEMU (Win-1)

PLAY WITH TECHNOLOGIES

Win-1

10.0.0.10

Win-2

15.0.0.10

Win-3

162.14.0.10

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]  
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>cd \

C:\>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=6ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>\_

8.1

8.2

Physical

01 : 00 : 00

START

28°

Windows Taskbar

freeSSHd update

Found New Hardware...

2 Windows Explorer

C:\WINDOWS\syst...

6:19 PM

ENG IN

22:49

21-08-2026

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QEMU (Win-2)

PLAY WITH TECHNOLOGIES

Win-1  
10.0.0.10

Win-2  
15.0.0.10

Win-3  
162.14.0.10

C:\WINDOWS\system32\cmd.exe

Pinging 10.0.0.10 with 32 bytes of data:  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 3ms, Average = 2ms  
C:\>ping 162.14.0.10  
Pinging 162.14.0.10 with 32 bytes of data:  
Reply from 162.14.0.10: bytes=32 time=4ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124  
Ping statistics for 162.14.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 4ms, Average = 2ms  
C:\>

8.1

8.2

162.14.0.10

Physical

01 : 00 : 00

START

28°

Start

freeSSHd update

C:\WINDOWS\s...

C:\WINDOWS\...

Network Connec...

Local Area Conn...

6:20 PM

ENG IN

22:50

21-08-2026

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QEMU (Win-3)

PLAY WITH TECHNOLOGIES

Win-1  
10.0.0.10

Win-2  
15.0.0.10

Win-3  
162.14.0.10

C:\WINDOWS\system32\cmd.exe

Pinging 10.0.0.10 with 32 bytes of data:  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 3ms, Average = 2ms  
C:\>ping 15.0.0.10  
Pinging 15.0.0.10 with 32 bytes of data:  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124  
Ping statistics for 15.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 3ms, Average = 2ms  
C:\>\_

Go

Physical

01 : 00 : 00

START

28°

Windows Taskbar

freeSSHd update Found New Har... C:\WINDOWS\... C:\WINDOWS\... 2 Windows Ex... 6:21 PM

ENG IN

22:51 21-08-2026



```
R1#
R1#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
-----
GigabitEthernet0/0       10.0.0.1        YES manual  up          up
GigabitEthernet0/1       20.0.0.1        YES manual  up          up
GigabitEthernet0/2       unassigned      YES TFTP    down        down
GigabitEthernet0/3       unassigned      YES TFTP    down        down
```

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
```

--More--

```
R2#
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
o0l
Ethernet0/0              20.0.0.2        YES manual up          up
Ethernet0/1              30.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
```

```
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
```

--More--

```
R3#sh ip int brief
```

| Interface   | IP-Address | OK? | Method | Status | Prot |
|-------------|------------|-----|--------|--------|------|
| Ethernet0/0 | 30.0.0.2   | YES | manual | up     | up   |
| Ethernet0/1 | 172.19.0.1 | YES | manual | up     | up   |
| Ethernet0/2 | 21.0.0.1   | YES | manual | up     | up   |
| Ethernet0/3 | unassigned | YES | TFTP   | down   | down |

```
R3#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, \* - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
```

--More--

```
R4#sh ip int brief
```

| Interface   | IP-Address | OK? | Method | Status | Prot |
|-------------|------------|-----|--------|--------|------|
| Ethernet0/0 | 15.0.0.1   | YES | manual | up     | up   |
| Ethernet0/1 | 172.19.0.2 | YES | manual | up     | up   |
| Ethernet0/2 | unassigned | YES | TFTP   | down   | down |
| Ethernet0/3 | unassigned | YES | TFTP   | down   | down |

```
R4#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, \* - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
--More--
```



```
R5#sh ip int brief
```

| Interface   | IP-Address    | OK? | Method | Status | Prot |
|-------------|---------------|-----|--------|--------|------|
| Ethernet0/0 | 21.0.0.2      | YES | manual | up     | up   |
| Ethernet0/1 | 192.168.168.1 | YES | manual | up     | up   |
| Ethernet0/2 | unassigned    | YES | TFTP   | down   | down |
| Ethernet0/3 | unassigned    | YES | TFTP   | down   | down |

```
R5#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, \* - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/0
L        21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
--More--
```



```
sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
o
Ethernet0/0              162.14.0.1      YES manual up          up
Ethernet0/1              192.168.168.2   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
```

```
R6#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        162.14.0.0/16 is directly connected, Ethernet0/0
L        162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.2/32 is directly connected, Ethernet0/1
```

```
R6#
```